

ARSH SHIVAYACH

Senior Software Engineer | Technical Leadership | AI Automation & Systems Engineering

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Open to Remote (preferred) or Hybrid – Noida

PROFESSIONAL SUMMARY

Software engineer with 6+ years building performance-critical C++/Qt desktop systems, enterprise applications, and CAD software, including crash analysis, memory safety, and performance tuning on production codebases. I write technical specifications and architecture, then use AI coding agents to accelerate implementation, refactoring, and testing under structured review. I established and helped drive adoption of this specification-first, AI-assisted workflow at Graebert. My background combines deep C++/systems engineering with modern AI-assisted development and automation, and I'm looking for a Technical Lead or Engineering Manager role where I can own architecture and delivery for a team.

CORE TECHNICAL SKILLS

Core Professional Experience (6+ years, production)

Languages & Systems: C++, STL, OOP, Design Patterns, SQL, Shell Scripting

Desktop / CAD Engineering: Qt, ODA SDK, CMake, CAD, Geometry Processing, Performance Engineering, Crash Analysis, Memory Safety

Engineering Practice: Software Architecture, Technical Specifications, Debugging, Regression Prevention, Testing, Git, Perforce, Jira, CI/CD

AI-Assisted Engineering (applied in production role): Specification-first development, AI coding agents (Windsurf/Claude, GitHub Copilot, ChatGPT, Gemini), AI-assisted implementation, refactoring & debugging

Project / Self-Directed Experience

Web, Backend & Data: Python, TypeScript, JavaScript, React, Next.js, Node.js, FastAPI, PostgreSQL, Redis, AWS, Docker

AI & Automation Platforms: AI agent orchestration, OpenAI API, Claude API, workflow automation

PROFESSIONAL EXPERIENCE

Software Engineer | *Graebert GmbH*

2023 – Present • Remote

- **Systems engineering:** Develop and enhance performance-sensitive C++/Qt CAD software, working with the ODA SDK on geometry processing, UI behavior, and platform-level engineering.
- **Technical leadership:** Write technical specifications, requirements, and architectural guidelines that define how features get built, shifting the team from ad hoc implementation toward a structured, specification-first process.
- **AI-assisted engineering:** Apply AI coding agents (Windsurf/Claude Sonnet, GitHub Copilot, ChatGPT, Gemini, Codeium) for implementation, refactoring, and debugging inside that specification-first process, with structured review rather than ad hoc code generation.
- **Debugging & performance:** Diagnose crashes and performance issues in production CAD workflows through root-cause debugging and profiling, and work to prevent regressions before release.
- **Delivery:** Coordinate with QA to validate implementation against specifications and bring builds to a release-ready, regression-safe state.
- **Enablement:** Support onboarding of the team onto AI-assisted workflows, writing documentation and workflow guidelines other engineers use.

Senior Software Engineer | *Agiliad Technologies*

2022 – 2023 • India

- **Imaging systems:** Built C++ imaging workflows integrating Fiji ImageJ and pugiXML for automated processing, data extraction, and configurable imaging pipelines.
- **Modernization:** Modernized legacy modules and built automation utilities, improving maintainability and making it easier to add new features.
- **Quality & release:** Worked with QA and engineering on design reviews, debugging, and release validation to keep delivery regression-safe.

- **Enterprise delivery:** Delivered features, enhancements, and defect fixes in C++ and SQL-based enterprise applications against defined requirements and test criteria.
- **SDLC:** Worked across the full SDLC — requirement clarification, implementation, code review, CI, and coordinated release pipelines — in a large-scale enterprise environment.
- **Production support:** Supported production readiness and post-release stability through debugging, patch creation, and controlled incremental rollouts.

SELECTED ENGINEERING PROJECTS

Self-directed engineering projects built outside of full-time employment, demonstrating current capability in AI-assisted engineering, automation, and full-stack product development.

AI Software Factory — *Windsurf, Claude, GitHub Copilot, ChatGPT, Gemini*

Built an engineering automation workflow that converts natural-language requirements into executable specifications and architecture, then uses AI agents to implement against those specifications with structured review and validation before acceptance.

AI Workflow Automation Platform — *Python, TypeScript, React, OpenAI API, Claude API*

Built an automation platform that orchestrates AI agents against backend and API integrations to run multi-step business workflows end-to-end, replacing manual coordination across tools.

Legal Work Infrastructure — *TypeScript, Node.js, React, PostgreSQL, AWS*

Designed the application architecture, database architecture, and backend for a legal document processing and workflow management system, including cloud deployment on AWS.

Developer Productivity Tools — *TypeScript, Node.js, Shell, Docker, GitHub Actions*

Built internal tooling and utilities to automate repetitive parts of the development cycle and support code quality checks in CI.

Enterprise Desktop / CAD Systems — *C++, Qt, STL, OpenGL, CMake*

Performance-critical desktop application work involving geometry processing and systems-level engineering, reinforcing the production C++/Qt foundation behind the AI and full-stack projects above.

EDUCATION

PG Diploma in Advanced Computing — CDAC - ACTS 2020

B.Tech, Electronics & Communication Engineering — Chandra Shekhar Azad University, Kanpur 2015 – 2019